AquaSense Pro

Water Quality Prediction and Aquaculture IoT ML Project
Round 2 Prototype Submission Report

Project Objective

This project develops an aquaculture water-quality and purifier monitoring system that combines IoT telemetry, sensor validation, and predictive analytics to improve operational visibility and decision support for end users.

The solution focuses on real-time monitoring of water parameters, process health, and purifier performance while remaining honest about unavailable hardware readings and simulated demo data.

The system is designed to operate in three data modes: live USB telemetry, CSV upload, and simulation scenarios. This makes the dashboard usable for testing, demonstration, and validation without forcing false sensor values.

Problem and Solution Summary

Aquaculture operations depend on stable water quality, filter performance, and process continuity. Poor data quality or missing telemetry can lead to incorrect interpretation of system health and delayed intervention.

The project addresses that issue by validating sensor inputs, normalizing telemetry, and clearly identifying when channels are missing or invalid. It also warns the user when a sensor is missing, faulty, or out of range.

In live USB mode, the app waits for real readings and intentionally does not fabricate values. In CSV and simulation modes, it demonstrates how the monitoring workflow behaves with realistic telemetry while keeping the status transparent and evidence-based.

System Flow and Functional Scope

The dashboard uses a segmented data-source interface for Live USB, Upload CSV, and Simulation. The live path checks for connected serial devices and reports a clear no-device state when no ESP32 is available.

The CSV path allows project users to upload telemetry, preview a bundled sample, or use a generated template. Simulation mode provides demo scenarios such as nominal operation, pressure trip, sensor fault, and flow imbalance.

Telemetry is normalized into metrics such as pH, temperature, turbidity, pressure, flow, TDS rejection, RO recovery, and mass-balance alarms. These outputs are displayed with plots and metrics only when the corresponding data channels are available.

Hardware and Validation Notes

The telemetry pipeline was tested with real device-compatible CSV content and simulated examples. The project documents that live hardware validation requires an actual connected ESP32, and the application never claims to measure a physical process when the board is absent.

This is a critical design decision to preserve trust and avoid false readings. The app warns the user when calibration or hardware validation remains pending, which is a responsible

engineering approach for a prototype system.

The project includes tests for telemetry parsing, alarm generation, and sample-data handling. This gives the current dashboard a dependable baseline and makes it easier to extend toward future hardware integration.

Submission Summary

Project name: AquaSense Pro

Project type: Water Quality Prediction and Aquaculture IoT ML

Primary mode: Streamlit dashboard with live USB, CSV upload, and simulation support

Delivery status: Completed prototype with functional demo workflow and validation tests

Important note: The dashboard clearly distinguishes between measured hardware data and simulated data.

Prepared for review and further development during the next round of assessment.